

What's New in Audio and Video Components



Abhimanyu Rathore is a content specialist at EFY



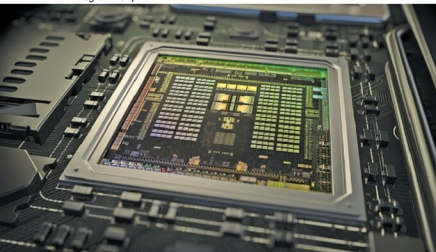
Chandramauli Mitra is an electronics circuit and chip design enthusiast

What if you could hear everything said in a far-off room that is also sealed behind sound-proof glass by merely recording a video of some small object over there, even if it were just the leaves of a potted plant? That is how advanced audio and video processing hardware have come to be. Researchers from MIT, Microsoft and Adobe have been able to recover intelligible speech from vibrations of an object by using a high-speed camera, whose video chips can process up to 6000 frames per second (fps). Of course, industry-leading devices can do up to 100,000fps. Compare that to a humble camera that can barely keep up with 60fps and you can get the picture (pun intended!) of how audio and video processing solutions have advanced over the ages.

Moore's law is taking everyone along

It was not that far back when Moore's law citations were limited to Intel processor launches. The one lasting trend in electronics is integration, and its increasing levels bring in different functionalities into the chip. What used to be discrete components that were soldered all over a PCB around five years back are now coming within a single piece of silicon.

NVIDIA Tegra X1 chip



If you get a pair of high-end headphones of the brand Beats by Dr Dre and open these up, there is just one chip inside. This one chip takes care of the entire sound quality and the experience that you have when you listen to music. It is a very strong audio platform that also streams Bluetooth from your iPod or your phone.

"What we see now is a single chip, a piece of silicon that has audio codec (coder-decoder), memory, processor and power management, all rolled into one. This is like a complete voice and music platform. This is a market that goes into headsets, sound bars and speaker docks, and the next big opportunity is to have the same system but stream it into the entire house using a combination of Bluetooth and Wi-Fi," explained Joep A.J. van Beurden, chief executive officer of CSR PLC, in an interview with EFY.

Chips that provide many-in-one functionalities such as dual microphones, single-chip mono-headset solutions with advanced echo and bi-directional noise-cancellation technology are targeted for low-cost two-microphone mono-headset applications provided by CSR's CSR8620.

Broadcom BCM2047 is an older monolithic 0.13µm, single-chip Bluetooth 2.1 + EDR wireless audio solution with integrated DSP. It offers what the company claims to be the highest level of functional integration to deliver a cost-optimised bill of materials (BOM) via a flexible platform for headset development.

Let us shoot the video first

PC gaming and mobile devices. A graphic display often plays an important role when buying a smartphone or a gaming PC. A serious gamer knows some of the most complex lighting and graphic challenges in visual computing. These days, lifelike graphics is the most demanding feature for